

Life Science > Core Bioreagents > Biological Buffers > Learning Center > Buffer Reference Center

Life Science Home

Life Science Products

ADME/Tox

Cell Biology

Cells and Cell-Based Assays

Cell Culture

Core Bioreagents

Learning Center

Biological Buffers

Learning Center

Buffer Calculator

Buffer Reference Center

Biological Buffer Products

BIS-TRIS

BIS-TRIS Propane

HEPES

HEPES Sodium Salt

MES

MES Sodium Salt

MOPS

MOPS Sodium Salt

Sodium Chloride

Trizma® HCl

Trizma®

Water

S-A BPC Advantage

Biological Buffers Quality System

Convenience Packaging

Custom Oligos

Epigenetics

Functional Genomics & RNAi

Gene Search

Metabolomics

Molecular Biology

Nutrition Research

Proteomics

Roche Biochemical Reagents

Stem Cell Biology

Synthetic Biology

Translational Research Solutions

Zinc Finger Nuclease (ZFN)

Learning Center

Labware

Biological Buffers

Buffer Reference Center

Useful pH Ranges of Selected Biological Buffers (25 °C, 0.1M)

- Useful pH Ranges of Selected Biological Buffers (25 °C, 0.1M)
- Trizma Buffer Table – pH vs. Temperature
- Phosphate Buffer Table – 0.2M solution
- Citric Acid – Na₂HPO₄ Buffer Solutions, pH approx. 2.6–7.6
- Citric Acid – Sodium Citrate Buffer Solutions, pH 3.0–6.2
- Sodium Acetate – Acetic Acid Buffer Solutions, pH 3.7–5.6
- Na₂HPO₄ – NaH₂PO₄ Buffer Solutions, pH 5.8–8.0 at 25 °C
- Imidazole (glyoxaline) – HCl buffer solutions, pH 6.2–7.8 at 25 °C
- Sodium Carbonate – Sodium Bicarbonate Buffer Solutions, pH 9.2–10.8
- Helpful Formulas

Useful pH Ranges of Selected Biological Buffers (25 °C, 0.1M)

Buffers	Useful pH Range	pKa (at 20)	pKa (at 25)	pKa (at 37)	Reagent Grade	BPC Grade	Ultra Grade
MES	5.5–6.7	6.16	6.10	5.97	M8250	M2933	M5287
Bis-Tris	5.8–7.2	n/a	6.50	6.36	B9754	B4429	B7535
ADA	6.0–7.2	6.65	6.59	6.46	A9883	n/a	A8074
aces	6.1–7.5	6.88	6.78	6.54	A9758	A3594	A7949
PIPES	6.1–7.5	6.80	6.76	6.66	P6757	P1851	P8203
MOPSO	6.2–7.6	n/a	6.90	6.75	M8389	n/a	n/a
Bis-Tris Propane	6.3–9.5	n/a	6.8, 9.0	n/a	B6755	B4679	B9410
BES	6.4–7.8	7.17	7.09	6.90	B9879	B4554	B6420
MOPS	6.5–7.9	7.28	7.20	7.02	M1254	M3183	M5162
TES	6.8–8.2	7.50	7.40	7.16	T1375	T5691	T6541
HEPES	6.8–8.2	7.55	7.48	7.31	H3375	H4034	H7273
DIPSO	7.0–8.2	n/a	7.60	7.35	D9648	n/a	D0306
MOBS	6.9–8.3	n/a	7.60	n/a	M3295	n/a	n/a
TAPSO	7.0–8.2	n/a	7.60	7.39	T9269	T5566	T0432
Trizma	7.0–9.0	8.20	8.06	7.72	T1503	T6066	T6791
HEPPSO	7.1–8.5	n/a	7.80	6.66	H3137	n/a	n/a
POPSO	7.2–8.5	n/a	7.80	7.63	P3405	n/a	P7088
TEA	7.3–8.3	n/a	7.80	n/a	T1377	n/a	n/a
EPPS	7.3–8.7	n/a	8.00	n/a	E9502	E0276	E1894
Tricine	7.4–8.8	8.16	8.05	7.80	T0377	T5816	T9784
Gly-Gly	7.5–8.9	n/a	8.20	n/a	G1002	G3915	G7278
Bicine	7.6–9.0	8.35	8.26	8.04	B3876	n/a	B8660
HEPBS	7.6–9.0	n/a	8.30	n/a	H6903	n/a	n/a
TAPS	7.7–9.1	8.49	8.40	8.18	T5130	T5316	T9659
AMPD	7.8–9.7	n/a	8.80	n/a	A9754	n/a	A9074
TABS	8.2–9.6	n/a	8.90	n/a	T1302	n/a	n/a
AMPSO	8.3–9.7	n/a	9.00	9.10	A6659	n/a	A7585
CHES	8.6–10.0	9.55	9.49	9.36	C2885	n/a	C8210
CAPSO	8.9–10.3	n/a	9.60	9.43	C2278	n/a	C8085
AMP	9.0–10.5	n/a	9.70	n/a	A9879	n/a	A9199
CAPS	9.7–11.1	10.56	10.40	10.02	C2632	n/a	C6070
CABS	10.0–11.4	n/a	10.70	n/a	C5580	n/a	n/a

Trizma Buffer Table – pH vs. Temperature

pH at Temperature			g/L for 0.05 M Solution	
5 °C	25 °C	37 °C	Trizma HCL	Trizma Base
7.76	7.20	6.91	7.02	0.67
7.89	7.30	7.02	6.85	0.80
7.97	7.40	7.12	6.61	0.97
8.07	7.50	7.22	6.35	1.18
8.18	7.60	7.30	6.06	1.39
8.26	7.70	7.40	5.72	1.66
8.37	7.80	7.52	5.32	1.97
8.48	7.90	7.62	4.88	2.30
8.58	8.00	7.71	4.44	2.65
8.68	8.10	7.80	4.02	2.97
8.78	8.20	7.91	3.54	3.34
8.88	8.30	8.01	3.07	3.70

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1/4

8.98	8.40	8.10	2.64	4.03
9.09	8.50	8.22	2.21	4.36
9.18	8.60	8.31	1.83	4.65
9.28	8.70	8.42	1.50	4.90
9.36	8.80	8.51	1.23	5.13
9.47	8.90	8.62	0.96	5.32
9.56	9.00	8.70	0.76	5.47

Phosphate Buffer Table – 0.2M solution

Potassium Phosphate Monobasic Anhydrous g/L	Sodium Phosphate Dibasic Heptahydrate g/L	23 °C pH	Potassium Phosphate Monobasic Anhydrous g/L	Sodium Phosphate Dibasic Heptahydrate g/L	23 °C pH
22.4	3.49	5.7	10.80	29.51	6.9
22.08	4.29	5.8	9.36	32.73	7.0
21.60	5.37	5.9	7.92	35.95	7.1
21.05	6.60	6.0	6.72	38.63	7.2
20.40	8.05	6.1	5.52	41.31	7.3
19.56	9.93	6.2	4.56	43.46	7.4
18.60	12.07	6.3	3.84	45.07	7.5
17.64	14.22	6.4	3.12	46.68	7.6
16.44	16.90	6.5	2.52	48.55	7.7
15.00	20.12	6.6	2.04	49.09	7.8
13.56	23.34	6.7	1.68	49.89	7.9
12.24	26.29	6.8	1.27	50.81	8.0

Citric Acid – Na₂HPO₄ Buffer Solutions, pH approx. 2.6–7.6¹

Citric acid monohydrate, C₆H₉O₇ • H₂O, M. wt. 210.14; 0.1M-solution contains 21.01 g/L Na₂HPO₄, M. wt. 141.98; 0.2M-solution contains 28.40 g/L, or Na₂HPO₄ • 2H₂O, M. wt. 178.05; 0.2M-solution contains 35.61 g/L.

x ml 0.1M-citric acid and y ml 0.2M-Na₂HPO₄ mixed.

pH	x ml 0.1M-citric acid	y ml 0.2-Na ₂ HPO ₄
2.6	89.10	10.90
2.8	84.15	15.85
3.0	79.45	20.55
3.2	75.30	24.70
3.4	71.50	28.50
3.6	67.80	32.20
3.8	64.50	35.50
4.0	61.45	38.55
4.2	58.60	41.40
4.4	55.90	44.10
4.6	53.25	46.75
4.8	50.70	49.30
5.0	48.50	51.50
5.2	46.40	53.60
5.4	44.25	55.75
5.6	42.00	58.00
5.8	39.55	60.45
6.0	36.85	63.15
6.2	33.90	66.10
6.4	30.75	69.25
6.6	27.25	72.75
6.8	22.75	77.25
7.0	17.65	82.35
7.2	13.05	86.95
7.4	9.15	90.85
7.6	6.35	93.65

Citric Acid – Sodium Citrate Buffer Solutions, pH 3.0–6.2¹

Citric acid monohydrate, C₆H₉O₇ • H₂O, M. wt. 210.14; 0.1M-solution contains 21.01 g/L Trisodium citrate dihydrate, C₆H₅O₇Na₃ • 2H₂O, M. wt. 294.12; 0.1M-solution contains 29.41 g/L.

x ml 0.1M-citric acid and y ml 0.1M-trisodium citrate mixed.

pH	x ml 0.1 M-citric acid	y ml 0.1M-trisodium citrate
3.0	82.0	18.0
3.2	77.5	22.5
3.4	73.0	27.0

3.6	68.5	31.5
3.8	63.5	36.5
4.0	59.0	41.0
4.2	54.0	46.0
4.4	49.5	50.5
4.6	44.5	55.5
4.8	40.0	60.0
5.0	35.0	65.0
5.2	30.5	69.5
5.4	25.5	74.5
5.6	21.0	79.0
5.8	16.0	84.0
6.0	11.5	88.5
6.2	8.0	92.0

Sodium Acetate – Acetic Acid Buffer Solutions, pH 3.7–5.6¹

Sodium acetate trihydrate, $\text{CH}_3\text{COONa} \cdot 3\text{H}_2\text{O}$, M. wt. 136.09; 0.2M-solution contains 27.22 g/l.

x ml 0.2M-NaOAc and y ml 0.2M-HOAc mixed.

pH, 18 °C	x ml 0.2M-NaOAc	y ml 0.2M-HOAc
3.7	10.0	90.0
3.8	12.0	88.0
4.0	18.0	82.0
4.2	26.5	73.5
4.4	37.0	63.0
4.6	49.0	51.0
4.8	59.0	41.0
5.0	70.0	30.0
5.2	79.0	21.0
5.4	86.0	14.0
5.6	91.0	9.0

 Na_2HPO_4 – NaH_2PO_4 Buffer Solutions, pH 5.8–8.0 at 25 °C¹

$\text{Na}_2\text{HPO}_4 \cdot 2\text{H}_2\text{O}$, M. wt. 178.05; 0.2M-solution contains 35.61 g/l. $\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$, M. wt. 358.22; 0.2M-solution contains 71.64 g/l. $\text{NaH}_2\text{PO}_4 \cdot \text{H}_2\text{O}$, M. wt. 138.01; 0.2M-solution contains 27.6 g/l. $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$, M. wt. 156.03; 0.2M-solution contains 31.21 g/l.

x ml 0.2M- Na_2HPO_4 , y ml 0.2M- NaH_2PO_4 ; diluted to 100 ml with H_2O .

pH, 25 °C	x ml 0.2M- Na_2HPO_4	y ml 0.2M- NaH_2PO_4
5.8	4.0	46.0
6.0	6.15	43.85
6.2	9.25	40.75
6.4	13.25	36.75
6.6	18.75	31.25
6.8	24.5	25.5
7.0	30.5	19.5
7.2	36.0	14.0
7.4	40.5	9.5
7.6	43.5	6.5
7.8	45.75	4.25
8.0	47.35	2.65

Imidazole (glyoxaline) – HCl buffer solutions, pH 6.2–7.8 at 25 °C¹

Imidazole, $\text{C}_3\text{H}_4\text{N}_2$, M. wt. 68.08.

25 ml 0.2M-imidazole (13.62 g/l), x ml 0.2M-HCl, diluted to 100 ml with H_2O .

pH, 25 °C	x ml 0.2M-HCl
6.2	21.45
6.4	19.9
6.6	17.75
6.8	15.2
7.0	12.15
7.2	9.3
7.4	6.8
7.6	4.65
7.8	3.0

Sodium Carbonate – Sodium Bicarbonate Buffer Solutions, pH 9.2–10.8¹

$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$, M. wt. 286.2; 0.1M-solution contains 28.62 g/l. NaHCO_3 , M. wt. 84.0; 0.1M-solution contains 8.40 g/l.

x ml 0.1M- Na_2CO_3 and y ml 0.1M- NaHCO_3 mixed.

pH		x ml 0.1M- Na_2CO_3	y ml 0.1M- NaHCO_3
20 °C	37 °C		
9.2	8.8	10	90
9.4	9.1	20	80
9.5	9.4	30	70
9.8	9.5	40	60
9.9	9.7	50	50
10.1	9.9	60	40
10.3	10.1	70	30
10.5	10.3	80	20
10.8	10.6	90	10

Helpful Formulas**Percentage by weight (w/v)**

(% buffer desired / 100) x final buffer volume (ml) = g of starting material needed.

Molar Solutions

desired molarity x formula weight x solution final volume (L) = grams needed

Henderson-Hasselbach Equation

$$\text{pH} = \text{pKa} + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

Reference

1. Dawson, R. M. C.; Elliot, D. C.; Elliot, W. H.; Jones, K. M. *Data for Biochemical Research*; 3rd ed., Oxford Science Publ., 1986.

[back to top](#)

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